AI Messaging Sales Assistant

One AI assistant that answers customers on WhatsApp, Instagram and Messenger, books appointments, captures leads, and hands over to staff. Built for any business, with self-service sign-up.

Planning document, version 6

1. Goal

Answer every customer message within seconds, 24 hours a day, in Chinese and English. Capture leads, book appointments, and hand over to a person whenever one is needed. **The first version is a guided menu bot:** it asks, the customer taps an option, and the bot sends a prepared answer. It needs no AI and no AI fees. AI mode, which understands typed free text, is added later.

For	What it delivers
Customer	Instant replies in the chat app they already use, in their own language. A person is always one message away.
Staff	One inbox for all channels, a summary on every handover, and a simple way to give the chat back to the bot.
Business owner	A web admin panel to change prices, FAQ, the menu and bot behaviour without a developer, plus a daily summary on the phone.
Platform owner (you)	Businesses sign up and set up by themselves. You see usage and revenue, and can pause any business.

2. How it works

```
Customer (WhatsApp / Instagram / Messenger / website chat)
            |
            v
   Chatwoot inbox  --->  n8n (routes the message)
                             |
             staff already handling?  yes -> bot stays silent
                             | no
                             v
                       Bot service
   Guided menu (first version): sends the options, reads the
   customer's tap, sends the prepared answer, runs booking
   steps, saves the lead.
   AI mode (later): also understands typed free text.
                             |
                             v
            Reply sent back to the customer (seconds)

Needs a person? The customer taps Talk to staff, or types text the
menu cannot match twice: summary note, staff alerted, bot silent
until staff type /bot.
```

3. What the system includes

Module	What it does
Chat	One inbox for three channels. The bot answers first, staff take over at any time. Assignment, labels, private notes, canned replies.

Module	What it does
Menu bot (first version)	Guided questions the customer taps. A **menu builder** in the admin panel: messages and options in English and Chinese, answers with text, images or links, booking by taps (service, day, time, then name and phone), lead questions, talk to staff, and a fallback for typed text (show the menu again, then hand over). Starter menus per industry and usage statistics per option. WhatsApp allows up to 3 reply buttons or a list of 10 options per message (check current limits); the builder warns when a menu is too long.
AI mode (later, optional)	A switch that adds understanding of typed free text, suggested replies for staff, summaries, intent and mood. Off by default and paid per use.
Customers	Customer list with search, filters, tags, tiers, saved views, bulk actions and list settings. One customer ID across channels (matched by phone). Sub-pages: **Conversations** (every chat, including resolved, with search inside message text), **Tags** (customer tags and chat tags: create, rename, merge, colour), **Orders** (simple order or transaction records), **Review invitations** (who was invited, rating, whether they reviewed).
Bookings	Real availability from Google Calendar, **one calendar per practitioner or resource**. Book, reschedule, cancel, reminders.
Marketing	Template broadcasts and reactivation, only to customers who opted in.
Workflow	Automations (routing, reminders, follow-ups) edited visually in n8n.
Reports	Chats, leads by score, bookings, handovers, reply time, top questions.
Team	Staff accounts, roles, automatic assignment, routing by skill.
Settings	AI behaviour, catalog, resources, FAQ, hours, alerts, channel connections, plan and billing.

Smart extras

- **AI suggested replies** for staff, sent with one click.
- **Auto summary, tags, intent and mood** on every chat.
- **Translation view** so staff read and reply across Chinese and English.
- **Click-to-chat link and QR code** that also records where each customer came from.
- **After-hours mode**, **self-service rescheduling** in chat, and a **weekly owner insights** report.
- Later: aftercare and review requests, rebooking reminders, no-show recovery, waiting list, knowledge import from a price list or web page, review queue for unsure answers, deposit links.

More features for a complete product

Area	Features
Safety and control	**Approval mode** (AI drafts, staff approve; switch to automatic per channel or topic once trusted). Staff timeouts and escalation (holding message, reminder, reassign, alert the owner). Business hours and holiday rules. Consent, opt-out (STOP) and a customer blocklist.
Booking	Buffers between appointments, cancellation policy, deposits, packages and memberships (prepaid sessions), capacity and party size, no-show tracking, **multi-branch** (own hours, calendars and staff per branch).
Messaging	WhatsApp template manager (create, translate, approval status). Automatic translation of staff replies. Comment-to-DM for Instagram and Facebook. Website forms and chat widget.
Team	Tasks from a chat with assignee and due date. Quick-reply library and media library. Fine-grained roles and permissions, tag-based auto-assignment. Customer satisfaction survey. Customer list column settings.
Customer data	All-conversations list with filters and full-text chat search. Tag management for customer tags and chat tags. Orders list. Review invitation list with ratings. Media library with folders. Website form submissions list.
Automations	A template library with editable triggers and messages: welcome, reminders, post-visit survey, rebooking, no-show recovery, birthday, review request, hot-lead follow-up.
Platform	Meta setup wizard, quick-start checklist, help center and in-app support chat, feature requests, data import and export, API and webhooks, status page, branding per business.

Later: e-commerce and order integration, coupons and discount codes, more channels (TikTok, LINE, Telegram, email, WeChat), plugin store, security certifications, native mobile apps.

4. For every industry, with self-service sign-up

Building block	How it fits any business
Universal base	Replies, FAQ answers, lead capture and scoring, handover. Works even with no booking.
Catalog	Services or products with name, price and description in each language.
Booking types	Appointment with a resource, table, viewing, class, quote visit, or none.
Custom lead questions	The owner chooses what to ask (budget, party size, location and so on).
Starter templates	Clinic, dental, salon, real estate, restaurant, education, repair, retail and more. Pre-fill everything, fully editable.
AI-assisted setup	Drafts the setup from a description, website or price list. The owner approves before going live.

Sign-up journey

1	Sign up and verify email
2	Describe the business, or paste the website or price list
3	AI drafts the setup, owner reviews and approves
4	Connect Google Calendars and WhatsApp with guided steps and test buttons
5	Test chat, choose a plan, go live

5. Parts and tools

Part	Tool	Role
Channels	Meta WhatsApp, Instagram, Messenger APIs	Where customers write. Each business uses its own number and Meta account.
Inbox	Chatwoot (open source)	Staff inbox. One account per business, created automatically.
Automation	n8n (open source)	Routes messages, sends replies, handover alerts, reminders, reports.
Bot service	TypeScript API service in the aichat repo (+ Claude in AI mode)	Runs the menu flows, booking steps and lead capture. In AI mode it adds the prompt, business knowledge and tools.
Admin panel	Next.js web app (React)	Business settings, customers, bookings, reports. Platform-owner console. Theme colour and logo per business.
Calendar	Google Calendar API	Availability and bookings, one calendar per practitioner.
Database	PostgreSQL	All data, separated per business with row-level rules.
Sign-up and billing	Public site + Stripe or local provider	Registration, plans, usage limits, invoices.

6. Technical stack

All of our own code is TypeScript, in one repository, for both the web app and the server. Chatwoot and n8n run as ready-made Docker images, so they need no custom programming, only configuration.

Languages and frameworks

Part	Technology	Notes
Language and tooling	TypeScript (strict mode), Node.js 22, pnpm monorepo, ESLint	One codebase with shared types between the web app and the API.
API, AI and workers	Fastify (API), Zod (validation), BullMQ (background jobs on Redis)	Webhooks from Chatwoot and Meta, AI tool calls, booking, usage limits, reminders, billing events.

Part	Technology	Notes
AI model (later)	Claude through the Anthropic TypeScript SDK	Optional AI mode, added after the menu bot works. Default model Claude Haiku 5.5, set by one setting and changeable per business or plan.
Web app (admin, sign-up, platform console)	Next.js, React, Tailwind CSS, shadcn/ui components	Polished, interactive screens. Colours are theme variables, so each business can pick its own colour and logo, plus dark mode.
Inbox	Chatwoot (Ruby on Rails, Docker image)	Configured through its API and webhooks. One account per business, created through its platform interface.
Automation	n8n (Node.js, Docker image)	Workflows stored as JSON files in the repo and imported on start.
Calendar	Google Calendar API, googleapis library	Service account. Each business shares its calendars with it.
Billing and email	Stripe API (or a local provider), an email service over SMTP	Subscriptions, usage limits, invoices, webhooks. Verification emails and notifications.

Data, infrastructure and operations

Area	Technology	Notes
Database	PostgreSQL 16 with Drizzle ORM	SQL-first, works well with row-level security keyed by business ID. Separate databases for Chatwoot, n8n and our service on one server.
Cache and queues	Redis	Needed by Chatwoot. Also used for rate limits and background jobs.
Packaging and web	Docker Compose, Caddy	Whole stack in one file. Caddy gives automatic HTTPS. A public address is needed for Meta webhooks, through a tunnel such as ngrok during development.
Backups and monitoring	Scheduled pg_dump to cloud storage, Uptime Kuma, health checks in the admin dashboard	Per-business export and deletion. Alerts to Telegram or email.

Security and quality

Area	Approach
Login and secrets	Email and password with argon2 hashing, secure sessions, optional two-step login (TOTP) for owners. Meta tokens and Google credentials encrypted at rest; secrets in environment files, never in git.
Isolation and audit	Business ID on every table plus row-level security. Automated tests try to read across businesses. Sign-ins, support access, pauses and deletions are logged.
Testing	Vitest for logic (booking slots, handover rules, limits), Playwright for end-to-end screens, sample-message tests for workflows. GitHub Actions runs checks on every change.

Decided: TypeScript for all our own code. **Choices to confirm:** login library (Auth.js or Better Auth), Stripe or a local payment provider, and the email-sending service.

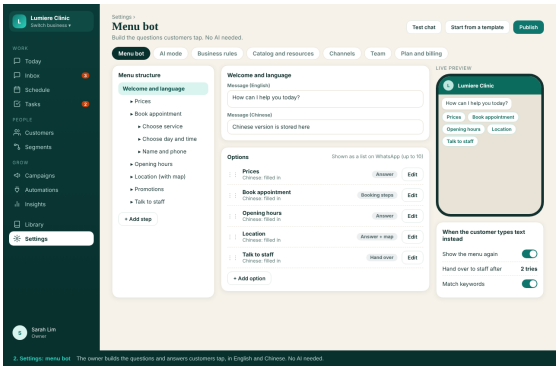
7. Design

A calm teal workspace, original to this product. The first version is menu-first, and AI features arrive with AI mode. Four of the most important screens are shown below. All 15 screens are in the separate page-layouts PDF.

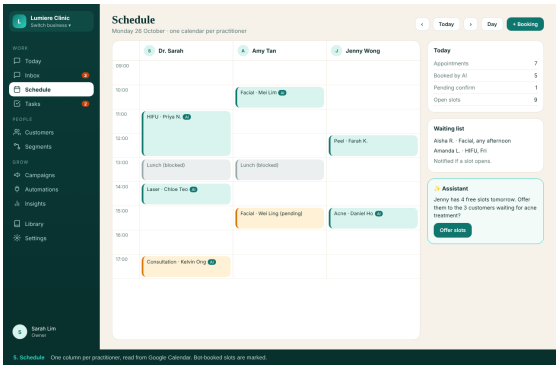
Element	Choice
Principles	Action first (home shows what needs a person), menu first and AI later (drafts, summaries and next steps appear once AI mode is on), cards over tables, plain wording.
Colours	Deep teal #08302d menu, teal #0f766e primary, mint #2dd4bf accent, cream #f7f5ef background. Status colours: hot, warm, cold.
Type and shape	Clean sans-serif interface, serif page titles, rounded cards, pill tabs.
Menu	WORK: Today, Inbox, Schedule, Tasks. PEOPLE: Customers (list, conversations, tags, orders, review invitations), Segments. GROW: Campaigns, Automations, Insights. Then Library and Settings.
Themes	Teal by default. Each business can pick its own accent colour and logo, and use dark mode.



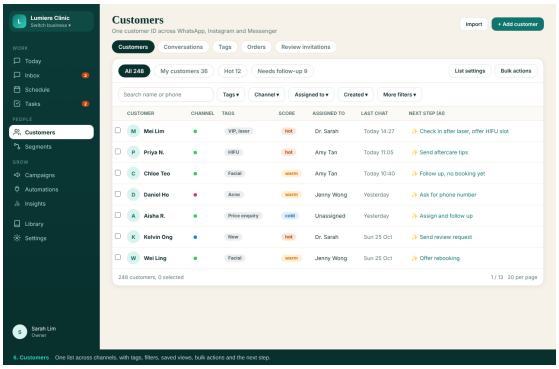
Customer chat: tap an option, prepared answer, booking by taps



Menu bot builder: questions and answers in English and Chinese, no AI



Schedule: one Google Calendar per practitioner



Customers: tags, filters, saved views, bulk actions

8. Handover, safety and separation

The bot hands over to staff when

- The customer taps Talk to staff or asks for a person.
- Complaint, anger, refund or payment problem.
- Medical, legal or safety concern. The AI never gives such advice.
- Typed text the menu cannot match after two tries (in AI mode: the question is outside the business's knowledge, or the AI is unsure).

Safety and separation

- Answers only from the business's own knowledge. Never invents prices, offers or availability.
- Tells customers it is a virtual assistant. Customer text is treated as untrusted.

- Every record carries a business ID, and database rules block cross-business reads.
- Encrypted credentials, two-step login for owners, audit logs, data export and deletion.
- Sign-up protection: email verification, bot check, limits, and a kill switch per business.
- If the AI service fails, the customer gets a polite message and staff are alerted.

9. Plans, hosting and costs

Item	Approach
Plans	Free trial, then paid monthly plans with limits on AI replies, staff and resources. At the limit the AI pauses, the owner is told, and messages still reach the inbox.
Hosting	Free hosting for development, the demo and an invite-only beta. Public sign-up needs a stable paid server with backups.
AI cost	None in the first version (menu bot). In AI mode: pay per use and small (Claude Haiku 5.5 by default), counted per business and passed on through plans.
Other costs	Meta fees are billed by Meta directly to each business. Also domain, email sending and payment provider fees.

10. Free-first testing and build phases

Everything in the first version can be built and tested for free, because the menu bot needs no AI. AI mode, added later, is the only paid part and testing it costs cents.

Part	How to test it for free
Servers	Docker on your own computer. No paid hosting while building.
Public address	A free tunnel (ngrok free account or Cloudflare quick tunnel) so Meta can reach your computer.
First chat channel	Chatwoot website chat widget in your browser. No Meta account needed.
WhatsApp	Meta's developer test number, to a few approved numbers, without business verification. Check Meta's current rules.
Instagram and Messenger	A Meta app in development mode works for people with a role on the app, so no app review while testing.
Google Calendar	Free API, a test calendar on a free Google account.
AI (later)	Not needed for the first version. In AI mode, a fake AI provider runs automated tests for free and real Claude is used for a few manual tests (about 100 test chats is roughly $0.50, my estimate). Needs an Anthropic API key with a little credit.

Local-first setup

Decision: build and test on localhost (your own computer) first. Going live is far away, so no server is needed yet.

Item	Plan
Runs on	Docker Desktop on your computer, 8 GB RAM recommended. One command starts the whole stack. Planned addresses: admin web app localhost:3000, API localhost:4000, inbox localhost:3001, workflow editor localhost:5678.
Meta and Stripe	Free tunnel to localhost: ngrok free fixed address or Tailscale Funnel (stable), or Cloudflare quick tunnel (changes on restart). Stripe's command-line tool forwards test payments.
Limits and moving	The computer must stay on and customers cannot reach localhost. For 24/7 testing, an Oracle Cloud Always Free server is an option (card check, availability varies). Paid hosting only at go-live (phase 7); the Docker stack moves as is.
How to run	Install Docker Desktop, clone the repo, copy .env.example to .env, run docker compose up, open the admin address.

Phase	What gets built	How it is tested	Cost
0. Test base	Docker stack on your computer (Chatwoot, n8n, PostgreSQL, Redis), tunnel, business ID on every table, website chat widget.	A message in the web widget gets a reply and shows in the inbox.	Free
1. Menu bot	Guided menu bot: menu builder, English and Chinese options, prepared answers with text, images and links, lead questions and scoring, talk to staff with summary note and alert, fallback for typed text, business hours, staff timeouts, consent and opt-out, usage statistics.	Tap through every menu path in the web chat in English and Chinese. Automated tests walk each path.	Free
2. Booking	Google Calendar availability, booking by taps and cancel, reminders, buffers, cancellation policy, deposits, capacity.	Book on a test calendar. A second booking for the same slot must fail.	Free
3. Business admin panel	Today, Inbox, Schedule, Customers (one ID per person), all conversations with full-text search, tag management, tasks, library (replies, media folders, forms), settings, roles and permissions.	Change a price or a menu answer in settings and see the bot use it. Test chat for every change.	Free
4. Real channels	WhatsApp test number (buttons and lists), Instagram and Messenger (quick replies) in development mode, template manager, Meta setup wizard.	Real messages with a few approved numbers.	Free
5. Automations and marketing	Automation templates, reminders with WhatsApp templates, satisfaction survey, rebooking, review request, opt-in broadcasts, review invitation list, orders list, reports, comment-to-DM.	Run each automation on test customers.	Free
5b. AI mode (optional)	Real Claude behind a provider setting: understands typed free text, suggested replies for staff (Copilot), summaries, intent and mood, approval mode for AI replies.	Fake AI for automated tests, then a few real chats to check quality.	Cents of AI
6. Many businesses and self-service	Tenant isolation, sign-up and email verification, AI-assisted setup, onboarding, plans and limits, billing, platform-owner console, abuse protection, legal pages.	Two test businesses cannot see each other's data. Full sign-up journey works.	Free (payment tests use the provider's test mode)
7. Beta readiness	Meta business verification, real WhatsApp number, paid server with backups and monitoring, domain, email sending, legal review. Invite-only beta with a few businesses.	Real customers in a small, watched group.	Paid hosting starts here
8. Public launch and growth	Open sign-up. Multi-branch, packages and memberships, e-commerce, more channels, remaining extras.	Gradual rollout with usage limits.	Paid

Phases 0 to 5 are free and need no AI. Phase 5b adds small pay-per-use AI costs. Paid hosting begins at phase 7, when real customers arrive.